

Chap 06. Condition Statement

Contents

01. if statement

02. switch statement

01 Conditions and If Statements

If ... Else

https://github.com/kkh3363/C_Lang

■ The if Statement

- Use the if statement to specify a block of code to be executed if a condition is true.

```
if (condition) {  
    // block of code to be executed if the  
    condition is true  
}
```

```
if (20 > 18) {  
    printf("20 is greater than 18");  
}
```

If ... Else

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■ Example

- In the example above we use two variables, x and y, to test whether x is greater than y (using the > operator). As x is 20 and y is 18, the condition is true, so the program prints "x is greater than y".

```
#include <stdio.h>

int main() {
    int x = 20;
    int y = 18;

    if (x > y) {
        printf("x is greater than y");
    }

    return 0;
}
```

If ... Else

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■ Example

- include <stdbool.h> when working with bool variables.

```
#include <stdio.h>
#include <stdbool.h>

int main() {
    int x = 20;
    int y = 18;

    bool isGreater = x > y;

    if (isGreater) {
        printf("x is greater than y");
    }

    return 0;
}
```

If ... Else

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■ The else Statement

- Use the else statement to specify a block of code to be executed if the condition is false.
- Syntax

```
if (condition) {  
    // block of code to be executed if the  
    condition is true  
} else {  
    // block of code to be executed if the  
    condition is false  
}
```

If ... Else

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■ Example else

- In the example above, time (20) is greater than 18, so the condition is false. Because of this, we move on to the else condition and print to the screen "Good evening". If time was less than 18, the program would print "Good day".

```
#include <stdio.h>

int main() {
    int time = 20;

    if (time < 18) {
        printf("Good day.");
    } else {
        printf("Good evening.");
    }

    return 0;
}
```


if Statement

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■ Using a Boolean Variable

- You can also store the condition in a boolean variable and use it in the if...else statement. This can make the code easier to read:

```
#include <stdio.h>
#include <stdbool.h>

int main() {
    int time = 20;

    bool isDay = time < 18;

    if (isDay) {
        printf("Good day.");
    } else {
        printf("Good evening.");
    }

    return 0;
}
```

The else if Statement

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■ else if Statement

- Use the else if statement to specify a new condition to test if the first condition is false.
- You can use else if to check multiple conditions, one after another.
- Syntax

```
if (condition1) {  
    // block of code to be executed if condition1  
    is true  
} else if (condition2) {  
    // block of code to be executed if condition1  
    is false and condition2 is true  
} else {  
    // block of code to be executed if both  
    conditions are false  
}
```

The else if Statement

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■ Example

- The value of time is 16. The first condition (time < 12) is false, but the second condition (time < 18) is true. Because of this, the code inside the else if block runs, and "Good day." is printed.
- If the value of time was 22, none of the conditions would be true, and the program would print "Good evening." instead.

```
#include <stdio.h>

int main() {
    int time = 16;

    if (time < 12) {
        printf("Good morning.");
    } else if (time < 18) {
        printf("Good day.");
    } else {
        printf("Good evening.");
    }

    return 0;
}
```

The else if Statement

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■ Example bool

- Just like with if statements, you can store conditions in boolean variables and use them with else if:

```
#include <stdio.h>
#include <stdbool.h>

int main() {
    int time = 16;

    bool isMorning = time < 12;
    bool isDay = time < 18;

    if (isMorning) {
        printf("Good morning.");
    } else if (isDay) {
        printf("Good day.");
    } else {
        printf("Good evening.");
    }

    return 0;
}
```

C Short Hand If Else

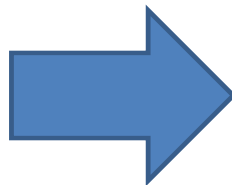
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■ Short Hand If...Else

- There is also a short-hand if...else, known as the ternary operator because it uses three operands.
- Syntax

```
variable = (condition) ? expressionTrue : expressionFalse;
```

```
int time = 20;  
if (time < 18) {  
    printf("Good day.");  
} else {  
    printf("Good evening.");  
}
```



```
int time = 20;  
(time < 18) ? printf("Good day.") :  
printf("Good evening.");
```

Nested If

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■ Nested If

- You can also place an if statement inside another if. This is called a nested if statement.
- A nested if lets you check for a condition only if another condition is already true.
- Syntax

```
if (condition1) {  
    // code to run if condition1 is true  
    if (condition2) {  
        // code to run if both condition1 and condition2 are  
true  
    }  
}
```

Logical Operators in Conditions

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■ Logical Operators in Conditions

- You can combine or reverse conditions using logical operators. These work together with if, else, and else if to build more complex decisions.
- && (AND) - all conditions must be true
- || (OR) - at least one condition must be true
- ! (NOT) - reverses a condition (true → false, false → true)

```
int a = 200;
int b = 33;
int c = 500;

if (a > b && c > a) {
    printf("Both conditions are true\n");
}
```

```
int a = 200;
int b = 33;
int c = 500;

if (a > b || a > c) {
    printf("At least one condition is true\n");
}
```

Nested If

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■ Example

- Find out if a person is old enough to vote, and if they are a citizen (using nested if statements):

```
#include <stdio.h>
#include <stdbool.h>

int main() {
    int age = 20;
    bool isCitizen = true;

    if (age >= 18) {
        printf("Old enough to vote.\n");

        if (isCitizen) {
            printf("And you are a citizen, so you can vote!\n");
        } else {
            printf("But you must be a citizen to vote.\n");
        }
    } else {
        printf("Not old enough to vote.\n");
    }

    return 0;
}
```


If ... Else

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■ Code Challenge

■ Instructions

- Inside main(), complete the following steps:
 1. Declare an int variable named age and assign it a value
 2. Use an if statement with the >= operator to check if age is 18 or older
 3. If the condition is true, print Allowed
 4. Otherwise, print Not allowed

```
#include <stdio.h>

int main() {
    // Write age here

    // Write an if statement that checks if age >= 18
    // If true, print "Allowed" inside the if block
    // Write an else statement
    // Inside the else block, print "Not allowed"

    return 0;
}
```

If ... Else

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■ Code Challenge

```
#include <stdio.h>

int main() {
    int age = 20;

    if (age >= 18) {
        printf("Allowed");
    } else {
        printf("Not allowed");
    }

    return 0;
}
```

Switch Statement

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- Instead of writing many if..else statements, you can use the switch statement.
- The switch statement selects one of many code blocks to be executed:
- Syntax

```
switch (expression) {  
    case x:  
        // code block  
        break;  
    case y:  
        // code block  
        break;  
    default:  
        // code block  
}
```

How it works:

- The switch expression is evaluated once
- The value of the expression is compared with the values of each case
- If there is a match, the associated block of code is executed
- The break statement breaks out of the switch block and stops the execution
- The default statement is optional, and specifies some code to run if there is no case match

Switch Statement

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■ Example

```
int day = 4;

switch (day) {
    case 1:
        printf("Monday");
        break;
    case 2:
        printf("Tuesday");
        break;
    case 3:
        printf("Wednesday");
        break;
    case 4:
        printf("Thursday");
        break;
    case 5:
        printf("Friday");
        break;
    case 6:
        printf("Saturday");
        break;
    case 7:
        printf("Sunday");
        break;
}
```

Switch Statement

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■ The break Keyword

- When C reaches a break keyword, it breaks out of the switch block.
- This will stop the execution of more code and case testing inside the block.
- When a match is found, and the job is done, it's time for a break. There is no need for more testing.

■ The default Keyword

- The default keyword specifies some code to run if there is no case match:

Switch Statement

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■ Example

```
#include <stdio.h>

int main() {
    int day = 4;

    switch (day) {
        case 6:
            printf("Today is Saturday");
            break;
        case 7:
            printf("Today is Sunday");
            break;
        default:
            printf("Looking forward to the Weekend");
    }

    return 0;
}
```

Switch Statement

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■ Challenge: Fix Customer Order

■ Instructions

- A customer wants to order Coffee.
- The code in the editor is incomplete. Fix it by inserting the missing values so the program prints the correct order.
 1. Change **choice** to the correct number for Coffee (see the comments).
 2. Insert **choice** inside **switch()**.
 3. Replace each **case:** with the correct number based on the comments above.
 4. Move **printf("Invalid choice\n")** into a **default:** case.

```
#include <stdio.h>

int main() {
    int choice = 2;

    // 1 = Coffee
    // 2 = Tea

    switch () {
        case:
            printf("You ordered Coffee");
            break;

        case:
            printf("You ordered Tea");
            break;

            printf("Invalid choice");
    }

    return 0;
}
```

Switch Statement

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■ Complete code

```
#include <stdio.h>

int main() {
    int choice = 1;

    // 1 = Coffee
    // 2 = Tea

    switch (choice) {
        case 1:
            printf("You ordered Coffee");
            break;

        case 2:
            printf("You ordered Tea");
            break;

        default:
            printf("Invalid choice");
    }

    return 0;
}
```


Thank You !